

31.65K

TOTAL TRIPS

£741.92K

TOTAL REVENUE

86.82%

ON-TIME RATE

13.18%

DISRUPTION RATE

£23.44

AVG TICKET PRICE

Project Overview

- **Scope:** Operational analysis of UK rail network across 10 major hubs.
- **Objective:** Transform ticket logs into actionable operational insights.
- **Dataset:** 31,653 transactions, 18 fields (Jan–Apr 2024 travel dates).

System Architecture

- **Data Pipeline:** CSV Raw Data → Power Query ETL → Star Schema Model.
- **Fact Table:** Central Fact Transactions table (31.65K records).
- **11 Dimension Tables:** Date, Stations, Ticket, Railcard, Delay Reason, etc.

Key Analytical Insights

- **First Class Upsell:** 9.7% of trips generate **20% of revenue** (£48.86 vs £20.72 avg).
- **Refund Driver:** Technical issues cause 472 trips but highest refund rate.
- **Peak Risk Window:** Demand spikes at 06:00 & 18:00 (3.1K trips), triggering delays.
- **Channel Mix:** Online (58.5% volume) vs Station (41.5% volume).

Tech Stack & Timeline

 Power BI

Power Query

</> DAX

 **Excel/CSV**

 GitHub

W1 Data Cleaning & Transformation

W2 Star Schema & DAX Measures

W3 Interactive Dashboard Design

W4 Forecasting & Final Report

Business Problem

- **Revenue Leakage:** £13.1K refund loss on Liverpool → Euston route alone.
- **Operational Blindspot:** Technical issues & signal failures lack root-cause visibility.
- **Reactive Planning:** Absence of time-series demand forecasting for resource allocation.

DAX Analytics Engine

Star Schema Modeling

Fact Table ↔ 11 Dimension Tables

- **Calculated Measures:** 17+ DAX measures (Total Revenue, Net Revenue, Refund Loss).
- **Performance Metrics:** On-Time Rate %, Delay Rate %, Cancellation Rate %.

May 2024 Demand Forecast

- **Time-Series Model:** Built-in Power BI Exponential Smoothing algorithm.
- **Projections:** Predicted daily ride volume and expected daily revenue for May 2024.

Actionable Decisions

- **Preventive Maintenance:** Focus on Liverpool-Euston technical reliability.
- **Off-Peak Incentives:** Smooth peak hours (06:00 & 18:00) using price discounts.